Beyond the Hype: What 18 Recent Research Papers Say about How to Use AI in Instructional Design

Aka, how to use (and how not to use) AI in your day to day work

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Hey folks!
The last couple of years have brought a relentless stream of AI promises to instructional design. Every conference, every LinkedIn post, every vendor pitch seems to suggest we're on the brink of a “professional transformation” that will make everything faster, cheaper and better.

Meanwhile, many of us are quietly concerned that AI might not be the 100% good news story we're being sold and that in practice the reality is... more complicated.

The last 18 months has seen a surge in controlled research on the impact of AI on the speed and quality of Instructional Design processes and outputs.

Thankfully, in the last 18 months amidst all of the hype we have seen a growing body of peer-reviewed research on AI & Instructional Design, providing us with systematic, robust and reliable data on how AI is being used in Instructional Design and how it’s impacting (for better or worse) the speed and quality of our work.

Perhaps unsurprisingly, the results are far more nuanced than either the AI evangelists or the skeptics would have you believe. Some findings might surprise you — others will validate concerns you've probably been too polite to voice in all those AI webinars you've attended recently.

TLDR:

The Efficiency Sweet Spot: AI excels at auto
lesson planning, 95% in assessment generati
and insight for quality and context.

The Creative Balance: AI can increase idea c
brainstorming partner, but over-reliance redu

The Skills Imperative: Designers who invest
better results—this is now a core professiona

Let's dive into what the research actually tells
should (and shouldn't) use AI in your day to d

Part I: Where AI Goes Wrong in Instructional Design

The research points to three critical risks that every instructional designer should understand before diving into AI adoption.

Risk #1: Pedagogical Blind Spots

The integration of AI into instructional design offers significant potential for efficiency and scalability. However, research consistently highlights critical risks associated with rigid, template-driven AI tools that constrain instructional creativity and reduce instructional designers' (IDs) agency. These tools often impose linear workflows that limit flexibility, reflective practice, and the ability to adapt lessons to diverse learner needs.

What the research reveals: Børte & Lillejord's study found that tools with rigid Instructional Design workflows significantly reduce user agency, making designers feel "boxed in" and less able and likely to tailor instruction to their students' needs. Researchers found that more flexible frameworks and interfaces like the ILUKS planner and ChatCLD, enable more free-flowing refinement of AI-generated learning designs across multiple pedagogical dimensions, improving contextual fit and user satisfaction.

Beyond workflow and creativity constraints, AI's fundamental pedagogical limitations pose a broader challenge. AI-generated content frequently lacks a deep understanding of learning science principles and struggles with contextual adaptation. Hu et al. discovered that 78% of GPT-4-generated math lesson plans required "significant adjustments" to align with local standards and learner backgrounds. Similarly, Krushinskaia et al. and Luo et al. observed that AI often reproduces instructional design language superficially without meaningful adaptation to specific learner needs.

Much of the research reports that AI's outputs frequently contain inaccuracies and fabricated information. Choi et al. and Madunić & Sovulj reported up to 40% fabricated references in AI-generated content, while DaCosta & Kinsell and Yıldızhan Bora & Kölemen documented regular cases of misinformation and bias. These issues underscore the necessity of rigorous fact-checking and expert review.

Over-reliance on AI can also erode professional judgment; Krushinskaia et al. found that 40% of designers accepted AI suggestions without adaptation, leading to less creative and contextually appropriate lessons. Meanwhile, Yang & Stefaniak highlighted concerns about deskilling and the need for more critical reflection on AI's role.

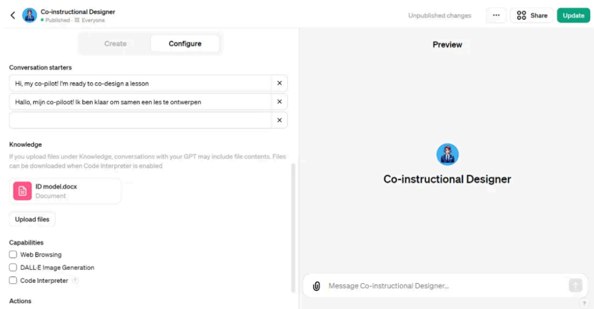


Figure 5: GPT Builder user interface. Example of a completed form in the "Configure" menu, part 3. Screenshot by the author.

The bot was made accessible to the public through the "Publish" button.

Overall, the process of bot creation was straightforward. The encountered difficulties involved the fact that all the materials presented to the bot must be short and concise; otherwise, the bot will ignore or misinterpret them. Another difficulty was situated in the nature of bot interaction: similar to how a human partner in instructional design would not completely replicate the wording used in two different cases, while relying on the same procedure, every interaction with the bot is unique and not completely predictable or controlled by the author. This can be attributed to ChatGPT's prior training, which every bot relies on. These largely unknown rules, often considered as a 'black box' to users, influence how any bot interacts. Additionally, a problem occurred when users did not follow the bot's lead and asked additional questions, such as "Provide me a lesson plan", before completing the required steps. This led to the bot replying to these requests and altering its pre-scripted procedure. To minimize the possibility of such cases, users were strongly advised to follow the bot's lead through the steps. Lastly, there were a small number of cases where, during interaction with the bot, it unexpectedly skipped steps in the ID model. The reason for this might be an overly extensive "Knowledge" file, which can be further shortened. This means that users must be informed about this possibility and should remind the bot to return to the correct order of steps if this happens.

5. User Feedback

Currently, the bot has been tested by 27 pre-service teachers studying in one of the Master's of Teaching programs at KU Leuven, Belgium. They were asked to design a two-hour lesson

Extract from Krushinskaia et al, showing their exploration of the risks and benefits of an AI "Co-instructional Designer".

To mitigate these risks, research advocates for treating AI outputs as drafts requiring expert validation and adaptation. Structured, human-centered frameworks like ChatCLD and ARCHED support iterative refinement and pedagogical alignment. Additionally, addressing equity and access challenges is vital to prevent widening learning gaps.

Regular reflection on AI's role and ongoing AI literacy training for all stakeholders ensure that AI remains a partner in instructional design rather than a crutch that diminishes expertise and learning quality.

TLDR:

- Avoid rigid, linear AI tools that limit design creativity and agency
- Use flexible, structured frameworks to refine AI outputs systematically
- Always fact-check and adapt AI-generated content; never accept it blindly

- Maintain human oversight to ensure nuance, context, and alignment with learning goals
- Address equity and access to prevent widening learning gaps
- Foster ongoing reflection and AI literacy to safeguard professional judgment and instructional quality

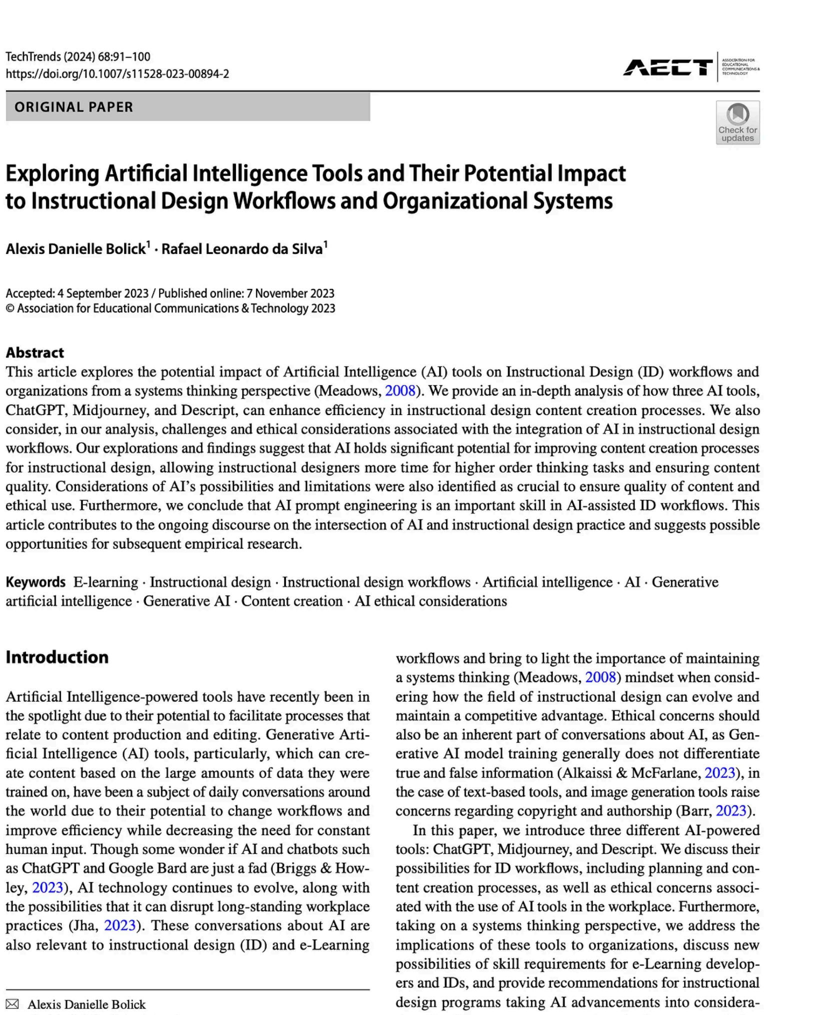
Risk #2: Ethics, Privacy & Bias

AI integration brings a host of ethical challenges that many instructional designers have yet to fully address. These risks are not theoretical: the research shows that issues of privacy, bias, copyright, and digital equity are already impacting learning environments.

What the research demonstrates:

Bias and Stereotyping: Bolick & da Silva's hands-on testing revealed some uncomfortable truths—AI tools perpetuated stereotypes in 28% of image outputs, raising concerns about the reinforcement of harmful biases in educational materials. DaCosta & Kinsell also noted that AI sometimes recommended media or delivery systems that were culturally or contextually inappropriate, highlighting the need for careful review.

Algorithmic Discrimination and Policy Gaps: Hodges & Kirschner's policy synthesis emphasised that AI-driven systems can amplify existing inequities and discrimination if not carefully monitored, especially when algorithms are trained on biased data sets or lack transparency.



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ORIGINAL PAPER

Exploring Artificial Intelligence Tools and Their Potential Impact to Instructional Design Workflows and Organizational Systems

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Abstract
This article explores the potential impact of Artificial Intelligence (AI) tools on Instructional Design (ID) workflows and organizations from a systems thinking perspective (Meadows, 2008). We provide an in-depth analysis of how three AI tools, ChatGPT, Midjourney, and Descript, can enhance efficiency in instructional design content creation processes. We also consider, in our analysis, challenges and ethical considerations associated with the integration of AI in instructional design workflows. Our explorations and findings suggest that AI holds significant potential for improving content creation processes for instructional design, allowing instructional designers more time for higher order thinking tasks and ensuring content quality. Considerations of AI's possibilities and limitations were also identified as crucial to ensure quality of content and ethical use. Furthermore, we conclude that AI prompt engineering is an important skill in AI-assisted ID workflows. This article contributes to the ongoing discourse on the intersection of AI and instructional design practice and suggests possible opportunities for subsequent empirical research.

Keywords E-learning · Instructional design · Instructional design workflows · Artificial intelligence · AI · Generative artificial intelligence · Generative AI · Content creation · AI ethical considerations

Introduction

Artificial Intelligence-powered tools have recently been in the spotlight due to their potential to facilitate processes that relate to content production and editing. Generative Artificial Intelligence (AI) tools, particularly, which can create content based on the large amounts of data they were trained on, have been a subject of daily conversations around the world due to their potential to change workflows and improve efficiency while decreasing the need for constant human input. Though some wonder if AI and chatbots such as ChatGPT and Google Bard are just a fad (Briggs & Howley, 2023), AI technology continues to evolve, along with the possibilities that it can disrupt long-standing workplace practices (Jha, 2023). These conversations about AI are also relevant to instructional design (ID) and e-Learning workflows and bring to light the importance of maintaining a systems thinking (Meadows, 2008) mindset when considering how the field of instructional design can evolve and maintain a competitive advantage. Ethical concerns should also be an inherent part of conversations about AI, as Generative AI model training generally does not differentiate true and false information (Alkaissi & McFarlane, 2023), in the case of text-based tools, and image generation tools raise concerns regarding copyright and authorship (Barr, 2023).

In this paper, we introduce three different AI-powered tools: ChatGPT, Midjourney, and Descript. We discuss their possibilities for ID workflows, including planning and content creation processes, as well as ethical concerns associated with the use of AI tools in the workplace. Furthermore, taking on a systems thinking perspective, we address the implications of these tools to organizations, discuss new possibilities of skill requirements for e-Learning developers and IDs, and provide recommendations for instructional design programs taking AI advancements into considera-

✉ Alexis Danielle Bolick

Bolick & da Silva's research on the impact of AI on Instructional Design Workflows (November, 2023)

Privacy and Consent Risks: The use of cloud-based AI tools introduces significant privacy risks. Bolick & da Silva identified that voice cloning and other generative AI features raised consent and data protection concerns. Sensitive learner data, if entered into cloud-based AI systems, could be exposed or misused. The ARCHED framework stressed the need for transparent, responsible and collaborative approaches to mitigate these risks.

Copyright and Intellectual Property: The proliferation of AI-generated media (images, text, audio) creates new copyright challenges for Instructional Designers. The research of Bolick & da Silva and Madunić & Sovulj separately highlighted the ambiguity around ownership and the importance of securing proper licenses for AI-generated assets.

Digital Equity and Access: Infrastructure barriers further complicate ethical AI adoption. Yıldızhan Bora & Kölemen documented how unreliable internet connectivity and limited device access prevented some students from fully engaging with AI-enhanced learning, effectively creating a two-tiered educational experience. This digital divide risks exacerbating existing inequities, particularly for marginalised learners.

Transparency and Stakeholder Communication: Studies such as ARCHED and ChatCLD emphasise the importance of being transparent with learners and stakeholders about the role, limitations, and risks of AI in instructional design. Clear communication helps manage expectations and fosters trust.

TLDR:

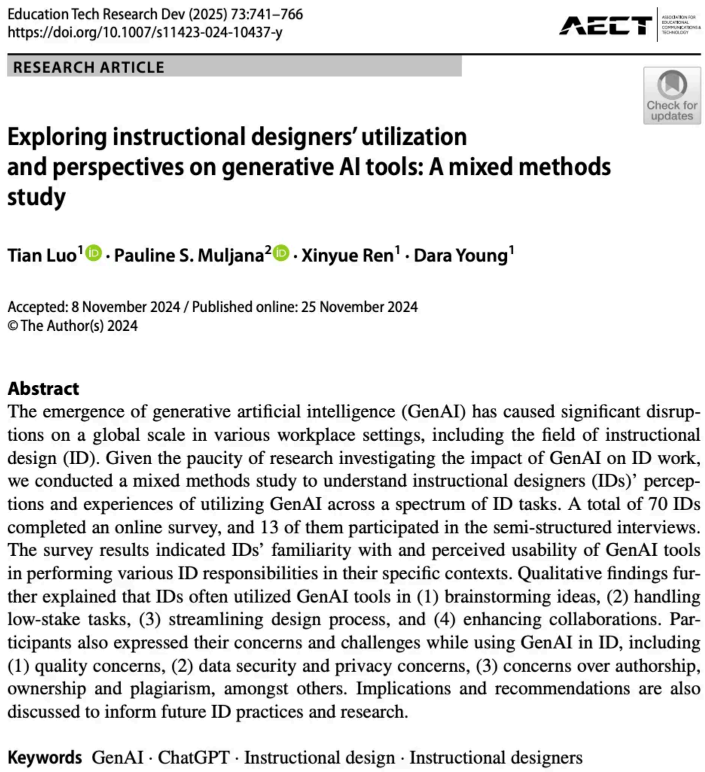
- Conduct regular bias audits and secure proper licenses for all AI-generated assets
- Never input sensitive learner data into cloud-based AI systems—use anonymised data or secure, on-premise solutions
- Proactively address digital equity by providing alternatives for learners with limited access
- Be transparent about AI's supplementary role and its limitations with all stakeholders
- Monitor evolving legal and policy frameworks to ensure compliance and ethical practice

Risk #3: The Creativity Trap

When it comes to creativity in Instructional Design, the research paints a nuanced picture. While AI can be a powerful catalyst for innovative thinking (more on this later), over-reliance on automated and standardised design suggestions may actually stifle the creative thinking and critical reflection that drive innovative learning experiences and deepen impact.

Research findings:

Diminished Creative Diversity: Luo et al. found that excessive AI use correlated with a 32% reduction in unique assessment designs among instructional designers. The convenience of AI-generated templates and suggestions can lead to over-standardisation, with fewer novel or contextually tailored solutions emerging in course development.



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RESEARCH ARTICLE

Exploring instructional designers' utilization and perspectives on generative AI tools: A mixed methods study

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Abstract
The emergence of generative artificial intelligence (GenAI) has caused significant disruptions on a global scale in various workplace settings, including the field of instructional design (ID). Given the paucity of research investigating the impact of GenAI on ID work, we conducted a mixed methods study to understand instructional designers (IDs)' perceptions and experiences of utilizing GenAI across a spectrum of ID tasks. A total of 70 IDs completed an online survey, and 13 of them participated in the semi-structured interviews. The survey results indicated IDs' familiarity with and perceived usability of GenAI tools in performing various ID responsibilities in their specific contexts. Qualitative findings further explained that IDs often utilized GenAI tools in (1) brainstorming ideas, (2) handling low-stake tasks, (3) streamlining design process, and (4) enhancing collaborations. Participants also expressed their concerns and challenges while using GenAI in ID, including (1) quality concerns, (2) data security and privacy concerns, (3) concerns over authorship, ownership and plagiarism, amongst others. Implications and recommendations are also discussed to inform future ID practices and research.

Keywords GenAI · ChatGPT · Instructional design · Instructional designers

Luo et. al (November, 2024)

Passive Consumption and Reduced Critical Thinking: Yıldızhan Bora & Kölemen's study of a digital photography course revealed that students who relied heavily on AI feedback reported diminished creativity and critical thinking. Instead of actively engaging with content, some learners became passive consumers of AI-generated suggestions, missing opportunities to develop their own ideas and problem-solving skills.

Risk of Homogenisation: Krushinskaia et al. observed that teachers who accepted AI-generated lesson suggestions without adaptation produced less creative and less contextually appropriate lessons. This trend risks homogenising instructional approaches and reducing the diversity of learning experiences.

Balancing AI and Human Ingenuity: Research by DaCosta & Kinsell and the ARCHED framework suggests that AI is most valuable as a brainstorming partner—expanding the solution space and providing raw material for human teams to refine. However, the creative process must be intentionally preserved through deliberate "human-only" design sessions and reflective practice.

Encouraging Reflection and Critique: Studies recommend requiring both students and designers to critique or build upon AI outputs, fostering metacognition and deeper learning. Regular audits of course materials for diversity of teaching approaches can help guard against over-standardisation and ensure a vibrant, creative learning environment.

TLDR:

- Balance AI use with deliberate "human-only" design sessions to preserve creativity
- Require personal reflection and creativity in assignments—encourage critique and adaptation of AI outputs
- Regularly audit courses for diversity of teaching approaches to avoid over-standardisation
- Use AI as a creative partner, not a replacement for human ingenuity and critical thinking

Part II: The Real Benefits of AI in Instructional Design

Understanding the risks of AI is crucial, but so too is understanding its real, measurable benefits. The same body of recent research that reveals AI's dangers also brings to light where it might genuinely transform instructional design for the better. Rather than dismissing AI wholesale or embracing it uncritically, the evidence points toward strategic integration—leveraging AI's strengths while maintaining human oversight in areas where it excels.

The research reveals four domains where AI consistently delivers value to instructional designers, often in ways that complement rather than compete with human expertise.

1. The Efficiency Revolution

AI is transforming instructional design by dramatically accelerating routine tasks—including assessment creation and lesson planning—while maintaining rigor and freeing up designers for higher-value, creative work.

Research evidence:

Dramatic Time Savings: Studies consistently report significant reductions in time spent on routine instructional design tasks. For example, Choi et al. found that ChatGPT reduced lesson planning time by 65%, while Cheng et al. reported that their TreeQuestion system cut MCQ generation time by 95%.

Scaling Assessment with Rigour: Cheng et al. demonstrated that AI-enabled assessment generation led to a 300% increase in assessment volume without sacrificing rigour, provided there was expert oversight. However, the research stressed that AI-generated assessments must be reviewed to ensure alignment with learner profiles, delivery modes, and cognitive objectives. Bloom's Taxonomy was highlighted as a useful structure for guiding AI prompts and ensuring assessments target the desired cognitive levels.

Rapid, Iterative Content Development: Dickey & Bejarano's GAIDE framework enabled fast, iterative development of course content and assessments, further supporting the efficiency gains reported across multiple studies.

Limits of Automation: While AI excels at drafting, assessment generation, and automating repetitive work, studies agree that creative, contextual, and strategic instructional design tasks still require substantial human input. Human oversight remains essential for ensuring quality, relevance, and alignment with learning goals.

TLDR:

- Use AI for initial drafting, assessment generation, and other routine tasks to automate repetitive work and scale output
- Maintain human oversight for quality control—AI is a support tool, not a replacement
- Structure AI prompts with frameworks like Bloom's Taxonomy to guide assessment design at appropriate cognitive levels
- Redirect the time saved to creative and strategic work that only humans can do effectively

2. Differentiation & Localisation at Scale

AI's ability to adapt design to specific learners and contexts represents one of its most promising applications, moving instructional design beyond "spray and pray" one-size-fits-all approaches toward tailored, needs-based learning mapped to specific personas, motivations and contexts.

Key research insights:

From Generic to Persona-Driven Design: Several studies highlight how AI empowers instructional designers to move away from generic, undifferentiated training and instead design learning experiences mapped to distinct learner personas. Madunić & Sovulj demonstrated that custom chatbots, when tailored to learner profiles, improved concept mastery by 29%—a result attributed to the chatbot's ability to adapt explanations and practice to individual needs, rather than relying on uniform content for all learners.

Adaptive Pathways and Real-Time Differentiation: Multiple studies report that AI-powered chatbots and adaptive learning systems can analyse learner interactions and performance data to dynamically adjust content, feedback, and assessments. This enables real-time differentiation—so learners receive support, challenges, and resources mapped to their current skill level, goals, and preferred learning modalities, rather than being forced through a rigid, uniform curriculum. This is especially effective for foundational or routine skills, freeing human experts to focus on more complex, context-sensitive instructional needs.

Frameworks for Transparent, Persona-Aligned Design: The ARCHED framework explicitly connects every AI-generated suggestion to pedagogical rationales, ensuring that adaptations are not only transparent but also mapped to specific learner objectives and contexts. This structured, multi-stage workflow helps IDs design with empathy and intentionality, using AI to generate, analyze, and refine learning objectives and assessments for distinct personas rather than defaulting to generic solutions.

Efficiency and Scalability Without Sacrificing Relevance: By automating content generation and analysis for different learner segments, AI enables organisations to scale personalised learning without multiplying design workload. This allows IDs to spend more time on strategic decisions—such as mapping content to personas' motivations, access needs, and prior knowledge—rather than on repetitive content creation.

Human Oversight Remains Essential: While AI can power much of the differentiation and persona-mapping at scale, studies consistently emphasise the need for human review and refinement—especially for advanced, sensitive, or highly contextual content. IDs must ensure that AI-driven pathways remain accurate, inclusive, and aligned with organisational and learner goals.

TLDR:

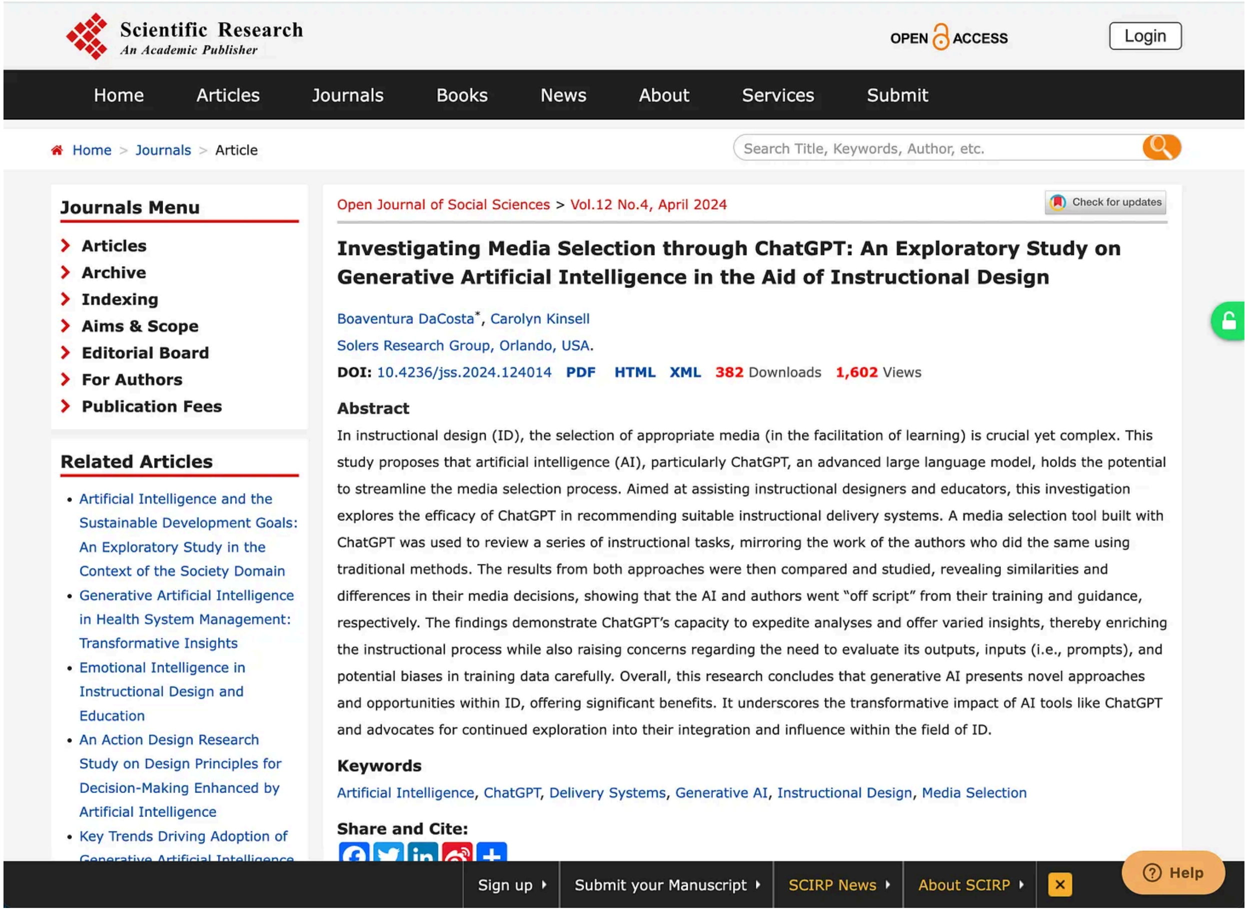
- Use AI to design learning mapped to specific needs, motivations, and contexts—not just generic content for the masses
- Implement adaptive learning pathways that adjust in real time to learner data, supporting differentiated instruction at scale
- Leverage frameworks like ARCHED to ensure every AI-driven adaptation is pedagogically justified and transparent
- Maintain human oversight for complex or sensitive content, but let AI handle routine differentiation and feedback

3. Amplified Creativity

AI is transforming instructional design by acting as a creative collaborator and expanding the diversity of design options available to teams. Rather than stifling creativity, research shows that AI can amplify it—serving as a brainstorming partner that rapidly generates a wide range of approaches, media, and activities that instructional designers might not have considered independently.

Evidence from the research:

Expanding the Solution Space: Luo et al. found that AI-assisted brainstorming increased idea diversity among instructional designers by 47%, demonstrating that AI can help teams break out of habitual patterns and explore new instructional strategies and formats. DaCosta & Kinsell's research further showed that ChatGPT-4 facilitated creative exploration of delivery systems and activity options, streamlining decision-making and offering diverse perspectives that enriched the design process.



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Investigating Media Selection through ChatGPT: An Exploratory Study on Generative Artificial Intelligence in the Aid of Instructional Design

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Solers Research Group, Orlando, USA.

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Abstract

In instructional design (ID), the selection of appropriate media (in the facilitation of learning) is crucial yet complex. This study proposes that artificial intelligence (AI), particularly ChatGPT, an advanced large language model, holds the potential to streamline the media selection process. Aimed at assisting instructional designers and educators, this investigation explores the efficacy of ChatGPT in recommending suitable instructional delivery systems. A media selection tool built with ChatGPT was used to review a series of instructional tasks, mirroring the work of the authors who did the same using traditional methods. The results from both approaches were then compared and studied, revealing similarities and differences in their media decisions, showing that the AI and authors went "off script" from their training and guidance, respectively. The findings demonstrate ChatGPT's capacity to expedite analyses and offer varied insights, thereby enriching the instructional process while also raising concerns regarding the need to evaluate its outputs, inputs (i.e., prompts), and potential biases in training data carefully. Overall, this research concludes that generative AI presents novel approaches and opportunities within ID, offering significant benefits. It underscores the transformative impact of AI tools like ChatGPT and advocates for continued exploration into their integration and influence within the field of ID.

Keywords

Artificial Intelligence, ChatGPT, Delivery Systems, Generative AI, Instructional Design, Media Selection

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DaCosta & Kinsell (April, 2024)

More Pathways, More Possibilities: The ARCHED framework study reported participants identifying 40% more design pathway options when using AI, underscoring AI's value in suggesting alternative approaches and media. This design diversity enables instructional designers to better accommodate a range of learning preferences and needs, supporting differentiation and inclusion.

Jumpstarting the Creative Process: Multiple studies and practical workshops highlight how AI can rapidly generate content variants, brainstorm course ideas, and outline different instructional materials, allowing teams to compare, combine, and refine options efficiently. This reduces cognitive fatigue and accelerates the creative process, freeing human designers to focus on higher-level strategic and contextual decisions.

Human Judgment Remains Essential: Across all studies, a consistent theme is that while AI excels at generating options, human expertise is critical for filtering, contextualising, and integrating these ideas into coherent, contextually appropriate solutions. AI's suggestions must be curated and adapted to fit the specific goals, audience, and constraints of each project.

TLDR:

- Initiate projects with AI brainstorming sessions to rapidly generate creative, diverse and innovative design ideas
- Use AI to explore multiple design pathways and media variants—expanding your repertoire beyond habitual choices
- Curate and adapt the best AI-generated ideas with team expertise; human judgment is essential for selection and contextual fit
- Leverage AI to accommodate diverse learning preferences and support inclusive, differentiated design
- Remember that AI's recommendations will not align with pedagogical best practices—check and validate everything

Conclusion: The Future of Instructional Design

Despite the hype about AI "revolutionising" education in general and Instructional Design in particular, the research tells a more nuanced story. The reassuring headline is that AI is not replacing Instructional Designers—it's augmenting their work in specific, measurable ways while simultaneously revealing what makes human expertise irreplaceable.

The efficiency gains afforded by AI free designers to focus on work that requires distinctly human capacities: ensuring pedagogical integrity and contextual adaptation demands the kind of nuanced judgment that emerges from expertise, experience and empathy. Ethical oversight and bias mitigation require moral reasoning that goes beyond algorithmic optimisation. Creative and strategic design depends on the ability to synthesise diverse perspectives and imagine possibilities that don't yet exist.

What we're seeing is Instructional Designers evolving into "augmented designers"—professionals who combine technological fluency with deepened human expertise. Their value increasingly lies in knowing when and how to leverage AI's capabilities while maintaining the critical distance necessary to evaluate, adapt, and contextualise its outputs.

A Skills Revolution for Instructional Design?

While many commentators have already hailed the death of prompt engineering as a key skill, the research consistently shows that designers who invest in prompt engineering and AI literacy achieve dramatically better results than those who approach AI casually. Krushinskaia et al. demonstrated that engineered prompts improved output relevance by 58% compared to basic requests. This reframes AI competency as a core professional skill rather than a technical add-on.

At its heart, Instructional Design has always been and remains a role which requires deep pedagogical knowledge, deep psychographic connection with learners, and creative problem-solving to address complex educational challenges. What has changed is not our role or purpose, but the toolkit that we have available for realising our goals.

The rise of AI empowers us to delegate the functional tasks which have distracted us for decades and return us to the true purpose of our role: deeply expert Instructional Design which drives learning outcomes.

The real question isn't whether AI will change instructional design—83% of us are already using it. The question is whether we'll use it thoughtfully to amplify rather than replace the expertise that makes great instructional design possible.

Happy experimenting!

Phil 👋

PS: Want to explore how to use AI through hands-on application with me and a cohort of people like you? Consider applying for a place on my [AI & Learning Design Bootcamp].

PPS: If you want to stay on top of the latest research, subscribe to my [Learning Research Digest] — a monthly digest of the most important research in instructional design.

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